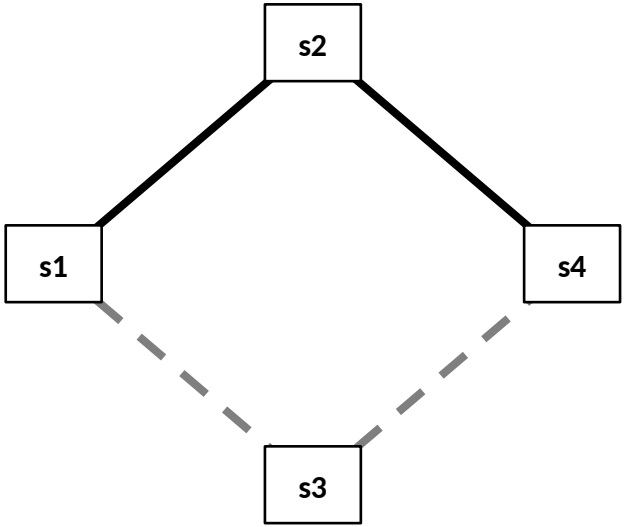
# SmartTE

## Centralized Traffic Engineering in Data Center Networks

BACSE203 Computer Networks, Fall 2026 | Topic 9

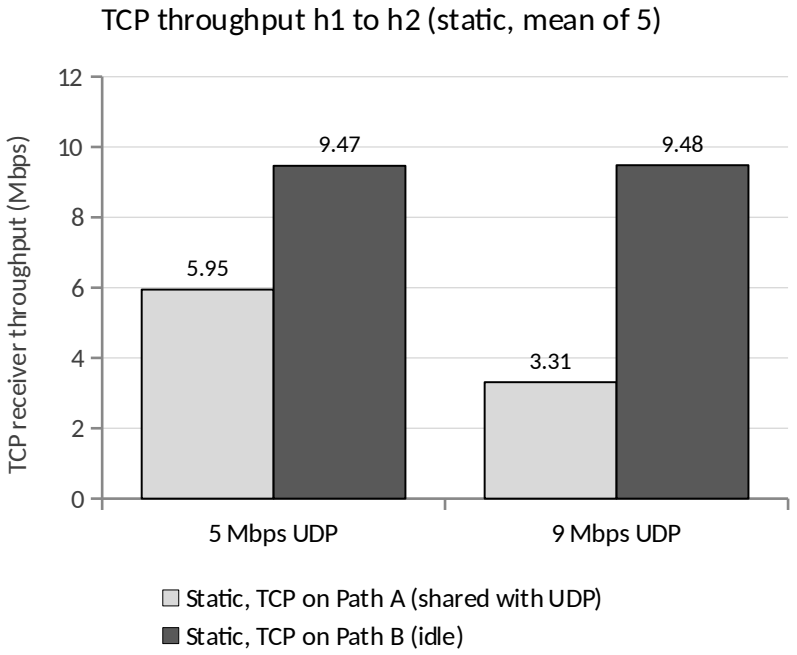
Aayush Santosh Singh 25BCE2341

Ashwin Krishna G 25BDS0038



# Problem: static paths waste path diversity

- Data-center fabrics offer multiple paths, but static forwarding can leave a flow on a congested one
- SDN lets a controller read link counters and rewrite flow rules over OpenFlow
- Open question: does reacting to utilization help, and what does it cost in route changes?




Source: report Table 1 (E2 static contrast, n = 5).

# Objectives and research question

Research question

When a controller scores paths by maximum link utilization, do a switching margin and a minimum dwell time reduce route oscillation, and what do they cost?

1

## Build

OS-Ken controller (OpenFlow 1.3) that reads port counters and installs flow rules

2

## Compare

Static, Policy B (lower utilization) and Policy C (margin + dwell)

3

## Evaluate

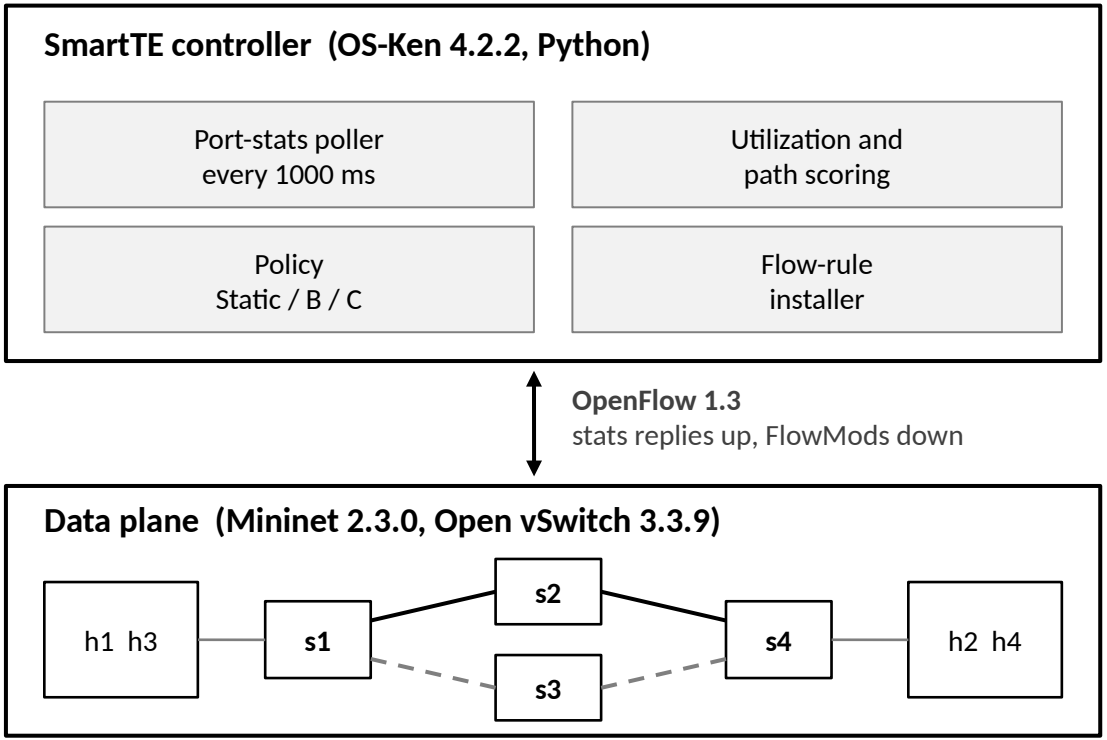
Four scenarios E1 to E4 in Mininet; throughput, RTT, loss, route events

4

## Report

Trade-offs honestly; characterize this implementation, not production TE

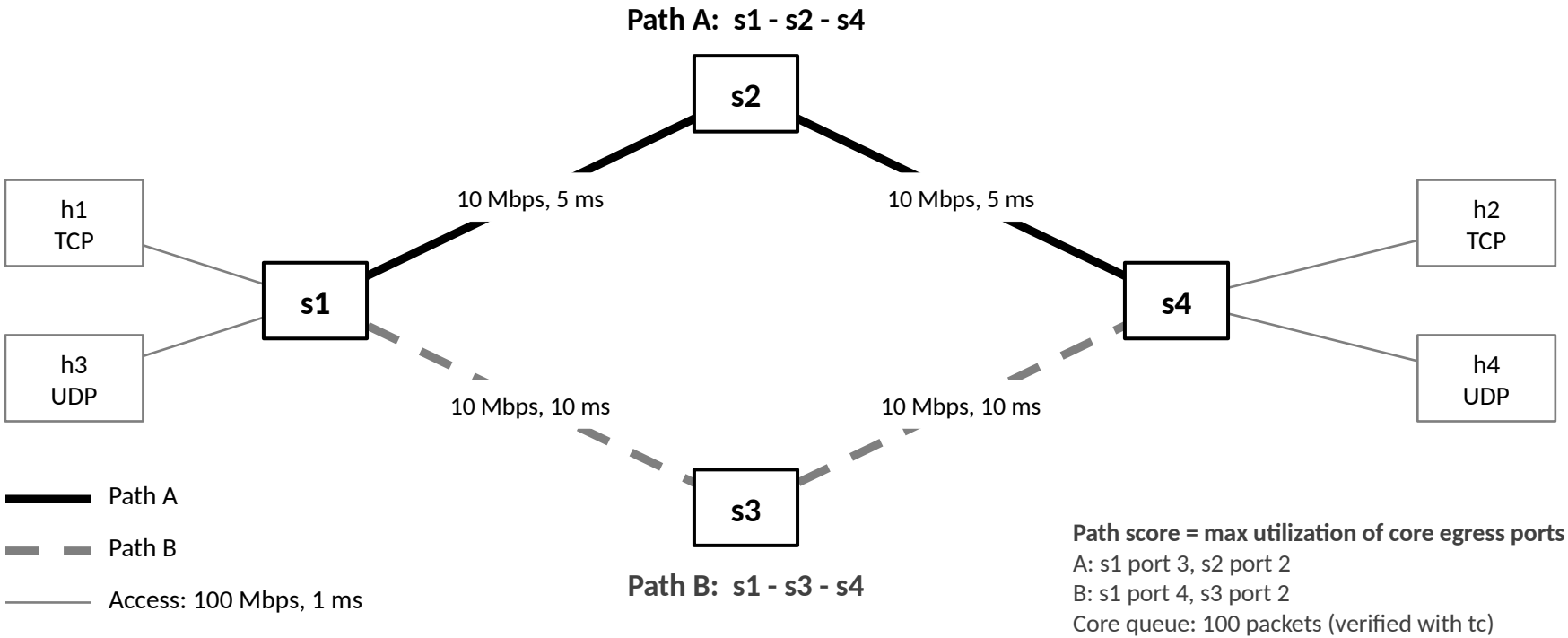
# System architecture




## Environment

- WSL2, Ubuntu 24.04.5 (aarch64)
- Mininet 2.3.0, OVS 3.3.9
- OS-Ken 4.2.2 (Python 3.10 env)
- iperf3 for TCP, ping for RTT/loss
- 20 unit tests pass (scoring, hysteresis, counters)

# Test topology: four hosts, four switches




**Path score = max utilization of core egress ports**
A: s1 port 3, s2 port 2
B: s1 port 4, s3 port 2
Core queue: 100 packets (verified with tc)

# Three forwarding policies compared

## Static

baseline

Keeps the configured path. Never makes a routing decision.

–

0 route changes

*Reference for what each path gives the TCP flow*

## Policy B

congestion-aware

Switches whenever the other path has a measurably lower utilization score.



16 route changes per trial (E1, no competitor)

*Reacts fast (about 0.2 s in E2)*

## Policy C

SmartTE: hysteresis + dwell

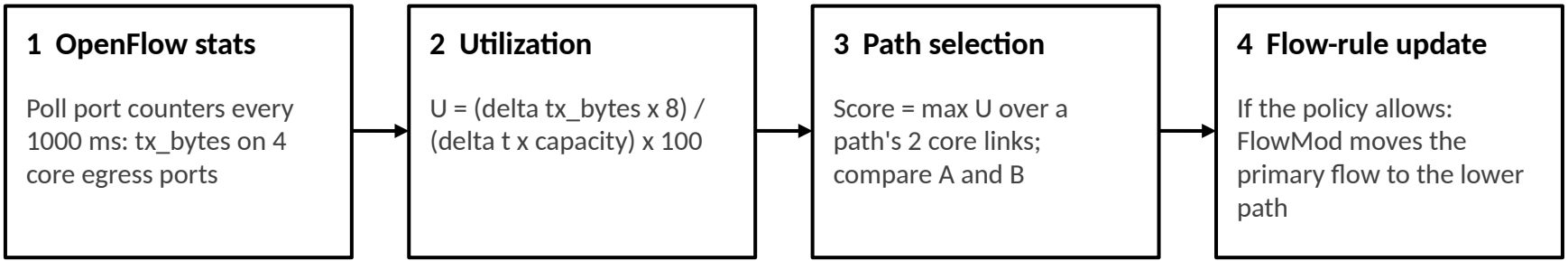
Also needs a 15 pp improvement and 5,000 ms since the last change.



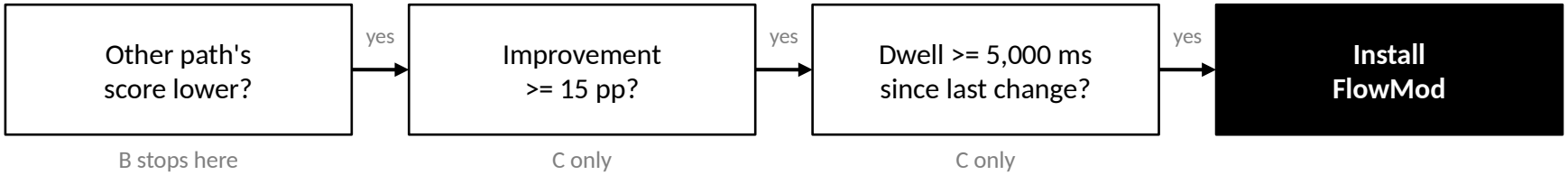
3 route changes per trial (E1, no competitor)

*Reacts slower (about 2.2 s in E2)*

# Controller workflow

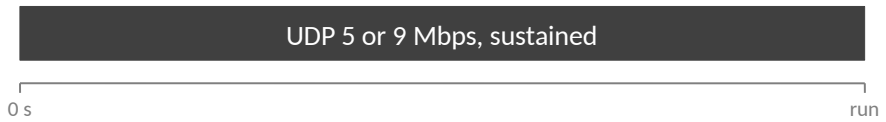
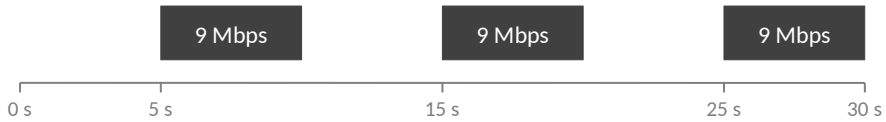



**Policy gate (decision at step 4)**



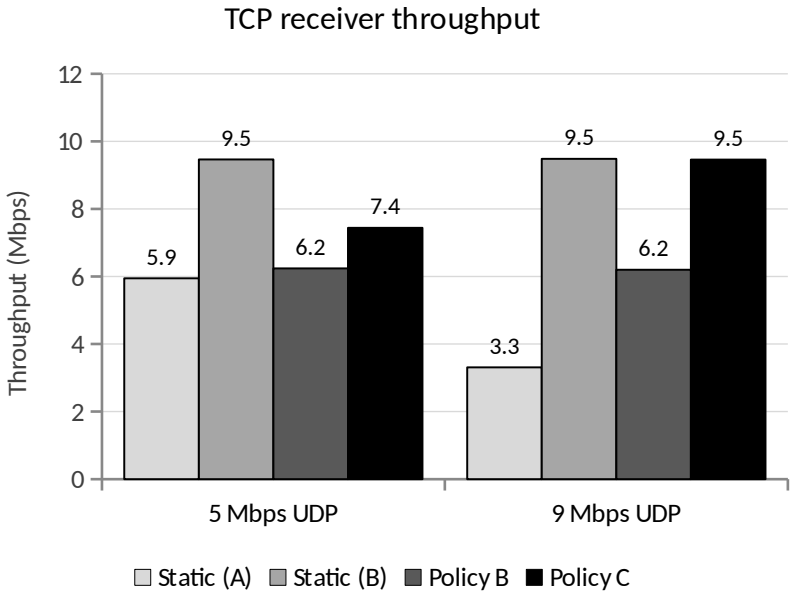

Any 'no' keeps the current path. Static never reaches the gate; B uses the first check only.

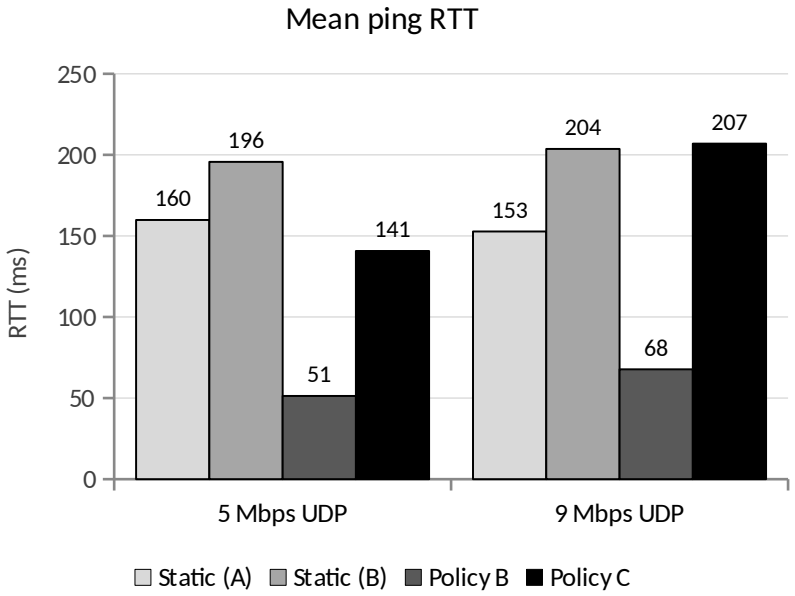
# Experimental setup: scenarios E1 to E4

| Scenario | | UDP load on Path A over time |
|---|---|---|
| **E1** | **No competing load**<br>TCP flow only (h1 to h2) |  |
| **E2** | **Sustained congestion**<br>UDP 5 or 9 Mbps on Path A; static A and B as references |  |
| **E3** | **Burst schedule**<br>Three 5 s bursts of 9 Mbps UDP in a 30 s run; C margin/dwell variants |  |
| **E4** | **Path A link failure**<br>Link s2-s4 disabled 1 s into a 15 s run |  |

Receiver-side iperf3 (TCP throughput), ping (RTT, loss), OpenFlow counters (utilization). 110 trials, n = 5 per cell, descriptive 95% intervals only.

# Throughput and latency under congestion (E2)

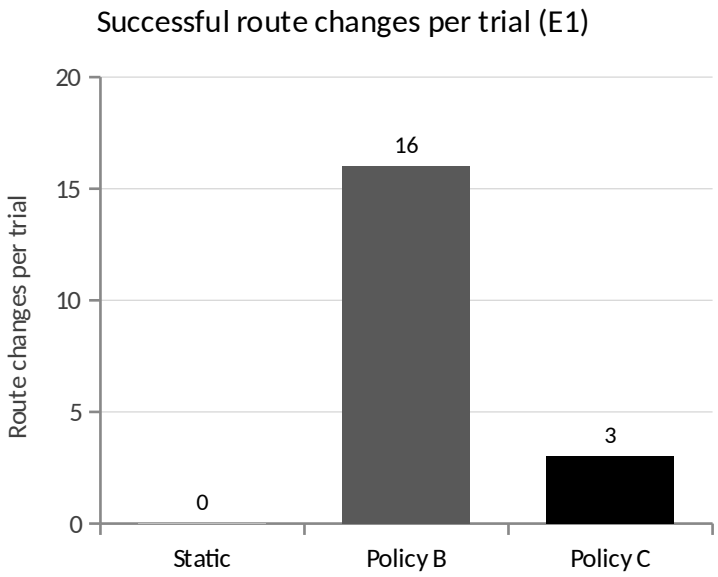



At 9 Mbps UDP, C reaches about 9.5 Mbps but at about 207 ms RTT; B has the lowest RTT (51 to 68 ms) but about 6.2 Mbps.

Static (A)/(B) values exact (Table 1); B and C read from report Figs. 3 and 5 (approx.). E1: Static about 9.5, B about 7.8, C about 9.3 Mbps. n = 5.

# Route stability and switching behaviour




Across E1 to E3: B made about 14 to 25 changes per trial, C about 2 to 9 (report Fig. 4).

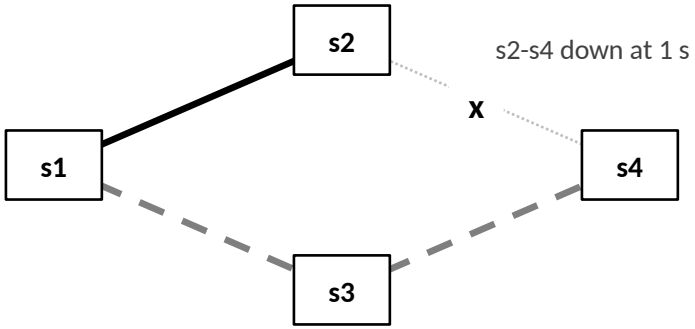
**E3: Policy C sensitivity (margin / dwell)**

| Margin / dwell | Changes per 30 s trial | TCP (Mbps) |
|---|---|---|
| 10 pp / 5 s | about 5 | about 9.1 |
| 15 pp / 2 s | about 8 | about 8.9 |
| **15 pp / 5 s (frozen)** | **about 5** | **about 8.9** |
| 15 pp / 10 s | about 3 | about 7.9 |
| 30 pp / 5 s | about 5 | about 8.9 |

Fewer changes can cost throughput: the largest dwell (10 s) switched least but lowered throughput. No best pair is claimed.

# Link failure (E4): diagnostic, not a ranking

| Policy | TCP results | On Path B at failure | Successful changes / trial | Attempts / rejected per trial | Recovery (s) | Ping loss (%) |
|---|---|---|---|---|---|---|
| **Static** | 0/5 | 0/5 | 0 | 0 / 0 | n/a | 93.33 |
| **Policy B** | 5/5 | 5/5 | 1 | 17 / 16 | 0.193 | 2.13 |
| **Policy C** | 5/5 | 0/5 | 1 | 11.8 / 10.8 | 1.005 | 1.87 |




- B had already moved to Path B before the failure; C was still on A and switched after it
- So E4 cannot rank failover latency fairly
- Raw decisions are not changes: every trial had 1 successful change, the rest were rejected
- Static stayed on the failed path: no TCP result in 5/5 trials

# Limitations and lessons learned

## Limitations

- Single-host Mininet/OVS on WSL2: timing, Linux queues and a recorded HTB warning can affect results
- One TCP flow, one UDP competitor, four switches, two paths
- n = 5 per cell; descriptive intervals, no significance test
- Git state was dirty (git_dirty = true), so the exact source is not fully pinned

## Lessons learned

- The controlled flow drives its own utilization signal: B switches even with no competitor
- Margin + dwell cut route changes but slow reaction (about 2.2 s vs 0.2 s)
- Count successful changes, not decisions: E4 raw 17 vs 1 successful
- Verify assumptions: queue limit of 100 packets checked with tc

# Conclusion

**5.9 to 9.5**

Mbps TCP, static on A vs B (5 Mbps UDP)

**16 vs 3**

route changes per trial, B vs C (E1)

**110**

trials, n = 5 per cell, 20/20 unit tests pass

- Path diversity helps only when assignment reacts to load
- A margin and dwell timer reduce route changes but delay action
- Evidence is specific to this emulation; no universal superiority is claimed

# Questions?

Aayush Santosh Singh 25BCE2341
Ashwin Krishna G 25BDS0038